

CS 230 : Discrete Computational Structures

Fall Semester, 2023

HOMEWORK ASSIGNMENT #5

Due Date: Thursday, October 12

Suggested Reading: Rosen Sections 9.1 and 9.5; Lehman et al. Chapter 10.5, 10.6 and 10.10

For the problems below, explain your answers and show your reasoning.

1. [10 Pts] For each of these relations decide whether it is reflexive, anti-reflexive, symmetric, anti-symmetric and transitive. Justify your answers. R_1 and R_2 are over the set of real numbers.

(a) $(x, y) \in R_1$ if and only if $xy = 4$

(b) $(x, y) \in R_2$ if and only if $x \geq 2y$

2. [5 Pts] Consider relation R_3 defined below on the set of positive real numbers. Prove that R_3 is a partial order, *i.e.*, it is reflexive, anti-symmetric and transitive.

$(x, y) \in R_3$ if and only if $x/y \in \mathbb{Z}$

3. [7 Pts] Consider relation R_4 defined below on the set of real numbers. Prove that R_4 is an equivalence relation. What is the equivalence class of 2? of π ? Describe all the equivalence classes using set-builder notation.

$(x, y) \in R_4$ if and only if $a = b + 3n$ for some integer n .

4. [6 Pts] Let R_5 be the relation on $\mathbb{Z}^+ \times \mathbb{Z}^+$ where $((a, b), (c, d)) \in R_5$ if and only if $a/c = b/d$.

(a) Prove that R_5 is an equivalence relation.

(b) Define a function f such that $f(a, b) = f(c, d)$ if and only if $((a, b), (c, d)) \in R_5$.

(c) Define the equivalence class containing $(1, 1)$.

(d) Describe the equivalence classes using set-builder notation.

5. [12 Pts] Prove that these relations on the set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$ are equivalence relations. Describe the equivalence classes.

(a) $R_6 = \{(f, g) \mid f(0) = g(0)\}$

(b) $R_7 = \{(f, g) \mid \exists C \in \mathbb{Z}, \forall x \in \mathbb{Z}, f(x) - g(x) = C\}$